

Activation and Foreclosure: Separating Screening from Exit Costs in Codified Moral Communities

Abstract

Strict communities are commonly explained by screening: costly requirements select committed members and deter free-riders. The same requirements also raise the cost of leaving, so observational evidence cannot separate selection from foreclosure. This paper separates them in an agent-based model of codified moral communities, in which agents observe conduct, sanction deviation, and decide whether to remain.

Three structural conditions, specified in advance rather than recovered from the runs, govern the outcome. An enforcement apparatus forms only where conduct is observable and enforcement is rewarded; the threshold is joint, and neither condition alone suffices. Punishment then concentrates in a small cadre, but that concentration is a product of institutional design rather than of differences among agents: the top five percent of active agents deliver 0.580 of all punishment under the full privilege architecture and 0.091 without it. An active apparatus retains members only where exit is foreclosed. With exit open, no run reaches capture in any of 63 parameter cells despite an activation rate of 0.598; imposed closure raises capture to 0.889. Where the apparatus does not activate, closing the exit produces nothing.

Exit foreclosure arises from external imposition or from internal absorption with generational replacement. The mechanism is defined over structural properties rather than doctrinal content, and applies to any organization that codifies conduct and attaches standing to compliance.

²³ **Keywords:** agent-based modeling; institutional design; norm enforcement; organizational capture;
²⁴ path dependence; religious organizations

1 Introduction

Why are strict religious groups often unusually cohesive and durable? The canonical answer begins with religious organizations as clubs that produce collective goods. In Iannaccone's account, apparently wasteful demands can make these clubs stronger: costly requirements involving dress, diet, ritual, time, and money screen out people whose low commitment would otherwise permit them to free-ride on the sacrifices of others (Iannaccone 1988, 1992, 1994). Those who remain redirect resources away from competing activities and toward the group, increasing both the average commitment of members and the value of membership itself. Strictness, on this view, is not organizational pathology but a solution to adverse selection and weak participation. This club-goods framework has become the dominant economic explanation of religious strictness, as reviews of the economics of religion emphasize (Iyer 2016; Carvalho 2019). It has also generated an unusually broad family of applications. Sacrifice and nontransferable religious capital help explain ultra-Orthodox Jewish education and fertility (Berman 2000); monitoring institutions can sustain religious production (McBride 2007); costly signals can resolve commitment problems in religious militancy (Berman and Laitin 2008); and veiling can operate as a commitment device that changes social interaction and marriage-market incentives (Carvalho 2013). Across these applications, costly observances select committed participants and align their investments with the group. This account identifies selection into strict groups, but it does not by itself identify ex post policing, the concentration of sanctioning authority, or retention when departure is costly. Research on punishment establishes that sanctions can sustain cooperation (Fehr and Gächter 2000; Fehr and Gächter 2002; Boyd et al. 2003, 2010; Hauert et al. 2007) and that metanorms can stabilize enforcement (Axelrod 1986). Most of these models make punishment costly to the punisher, whereas institutional enforcement can reward sanctioners with status, resources, and authority. The Emperor's Dilemma shows how observable enforcement and status rewards produce cascades (Centola et al. 2005), while Scott's account of legibility explains why standardized visible conduct is administratively tractable (Scott 1998). Neither contribution joins that activation mechanism to delegated authority and members' outside options.

Yet the same arrangements have another face. Mechanisms that screen at entry typically also raise the cost of exit after entry. A social network concentrated inside the group is valuable to a committed member, but it is also difficult to replace after departure. An identity made inseparable from membership can deepen solidarity, but it can also make departure psychologically and socially disorganizing. Distinctive dress, diet, language, schooling, and ritual can certify commitment among insiders while isolating members from outsiders who might provide employment, friendship, marriage, or recognition. Screening and exit foreclosure are therefore the same social structure described from opposite directions. Looking inward from the organizational boundary, strictness selects and coordinates participants; looking outward from the position of a member considering departure, it degrades alternatives and makes separation costly. Exit, voice, and loyalty identify departure as an organizational feedback mechanism (Hirschman 1970); preference falsification shows how public compliance can diverge from private belief (Kuran 1995). Work on leaving religion documents the social consequences of departure (Bromley 1998; Cragun and Hammer 2011; Streib et al. 2009), while models of secularization typically emphasize affiliation or cognitive change (Gelfand et al. 2011; Norenzayan and Gervais 2013). Case scholarship often recognizes screening and foreclosure at once without analytically separating them. An analysis of the Skoptsy sect, for example, argues that high entry costs screened for devoted individuals while high exit costs reduced the viable alternatives for living outside the sect (Engelstein 1999). The unresolved question is whether demanding systems persist because they select committed entrants or because they diminish viable exit. The two mechanisms can coexist, but coexistence does not establish which one accounts for durability.

That ambiguity is now a live theoretical problem rather than a terminological distinction. Bok (2025) observes that research on religious exit has concentrated mainly on who leaves, why they leave, and what follows, while the forms and roles of exit costs have received comparatively little systematic theorization. The systematic theorization of the exit-cost side is therefore recent relative to the mature literature on the determinants and consequences of exit. The institutional ingredients are otherwise familiar. Coercive monopoly and administrative penetration organize sanctioning ca-

capacity (Weber 1946; Tilly 1985; Mann 1986); institutions allocate enforcement power and generate increasing returns (North 1990; Greif 2006; Greif and Mokyr 2016; Pierson 2000); and histories of legitimating agents, hisbah, and inquisitorial jurisdiction show delegated enforcement in practice (Rubin 2017; Cook 2000; Given 1997; Kelly 1989). Pool-punishment models likewise distinguish delegated from peer sanctioning (Hilbe et al. 2014; Sigmund et al. 2010). The field has a mature screening account and a newly articulated exit-cost problem, but no design that can adjudicate their relative causal force. The identification problem is plain. In observational data, screening and exit foreclosure arrive bundled together. Any strict group that persists has already selected some people willing to bear its demands and has also organized relationships, identities, and opportunities in ways that make leaving consequential. Comparing strict and lenient groups cannot recover the separate contribution of either process, because the strictness that filters entrants also transforms the outside option of those who stay. Nor can studying former members solve the problem by itself: observed leavers are selected on having overcome the very costs whose preventive effect is at issue. Without independent variation, evidence consistent with successful screening is also consistent with blocked exit. The present synthesis makes code observability, enforcement reward, delegated privilege, and exit capacity independently variable. That separation yields entailments about when an apparatus activates, when it retains members, and why punishment concentrates.

Agent-based simulation supplies the missing instrument because it permits the two structures to be varied as distinct inputs while measuring any coupling between their effects. An experimental design can hold the observability and rewards that support enforcement constant while changing members' capacity to leave, or hold exit conditions constant while changing whether an enforcement apparatus can activate. No observational design can cleanly impose either intervention on otherwise equivalent religious communities. The strategy follows an established computational-religion program associated with Shults and Wildman: formal arguments for bringing simulation into the study of religion (Wildman and Sosis 2011), computational work linking cognition, culture, and social dynamics (Gore et al. 2018), and agent-based analysis in the economics of religion (Iannaccone and Makowsky 2007). Simulation here is therefore not a substitute for historical or

ethnographic evidence. It is a method for separating structural effects that those forms of evidence necessarily observe in combination.

This paper distinguishes two structural dimensions that are separable across most of the grid and coupled at the frontier margin: whether an enforcement apparatus activates, and whether members can leave. Four entailments of that specification organize the account. First, activation follows a joint threshold in code observability and the reward for enforcement; neither condition alone is sufficient. Second, institutional privilege concentrates punishment in a small cadre. Third, an active apparatus retains members only where exit is foreclosed. Fourth, foreclosure arises through external imposition or through internal absorption with generational replacement. Together they separate the conditions that form enforcement, retain its subjects, and concentrate coercive activity.

The concentration result also qualifies a prominent interpretation in the punishment literature. The costly-punishment tradition asks how sanctioning can arise through decentralized interaction even when punishers bear private costs. Experimental and evolutionary models show how punishment can sustain cooperation and how sanctioning behavior can spread or persist under specified population and institutional conditions (Fehr and Gächter 2000; Boyd et al. 2010; Hauert et al. 2007). From that perspective, a skewed distribution of punishment may look like an emergent product of heterogeneous strategies, repeated interaction, or selection among decentralized actors. The ablation suggests a different interpretation for delegated architectures. When one role enjoys exclusive or strongly asymmetric authority, lower costs and risks, and material reinforcement, concentrated sanctioning is substantially built into the opportunity structure before interaction generates any further inequality. This is not a general claim that punishment concentration never emerges from decentralized dynamics. It is a bounded contrast: in models with an institutionally privileged enforcement role, observed concentration cannot be treated as evidence of decentralized emergence without first removing or measuring the privileges that allocate coercive opportunities.

The mechanism is content-neutral. It is defined over structural properties: how observable a code's requirements are, what rewards attach to enforcing them, whether sanctioning authority is delegated asymmetrically, and how costly it is for a member to leave. Nothing in this definition

depends on the truth, theology, or substantive moral content of a doctrine. The same logic can apply to any organization that codifies conduct and attaches status to compliance, including political movements, residential communities, firms, schools, or ideological associations. Religious regimes provide the empirical terrain because they offer the longest and richest documented record of codified conduct, costly membership, delegated moral authority, and contested exit. They also make the duality especially visible, since practices experienced as identity, discipline, and mutual commitment by some members may function as barriers to outside life for others. The paper makes no claim that codified belief tends toward coercion. Its claim is conditional and structural: where observable rules, enforcement rewards, delegated privileges, and constrained exit combine, they generate dynamics that cannot be inferred from doctrine alone.

The remainder of the paper develops the argument in four steps. The next section defines code geometry and derives its structural regimes. The methods section specifies the agent-based design, the parameter-observable mapping, and the independent interventions on activation, exit, and privilege. The results section presents the activation frontier, exit-closure dose-response, and privilege ablation. The discussion states the empirical implications, limits, and tests that follow from separating selection into a group from members' capacity to leave it.

2 Theoretical Framework

2.1 Codified Moral Systems

A codified moral system is a socially shared normative structure in which obligations, prohibitions, and membership criteria are articulated in transmissible rule-form and maintained across generations through institutional mechanisms. Religious traditions are the most historically durable instances. The framework developed here is not restricted to religion; any identity-bearing system that codifies behavioral expectations and enforces compliance falls within scope. Political ideologies with loyalty tests, revolutionary movements with purity requirements, and professional bodies with ethical codes all instantiate codification to varying degrees.

As codified systems scale beyond face-to-face interaction, they confront a structural problem: moral quality must be made legible. Internal states (intention, sincerity, compassion, detachment) are opaque to observers. Institutions that must evaluate adherence therefore face pressure to adopt observable proxies. Ritual participation, dress codes, dietary restrictions, speech markers, formal declarations, and codified behaviors become instruments of classification. This shift is not incidental. It is a functional response to the problem of monitoring at scale. But it alters the incentive structure of the system. When moral standing becomes externally classifiable, enforcement roles become structurally viable.

This transformation is a problem of code geometry: the structural configuration of a codified moral system along dimensions that jointly determine whether enforcement equilibria become reachable.

2.2 Dimensions of Code Geometry

Five structural dimensions characterize the enforcement landscape of a codified system. These are not evaluative. They describe the formal properties of the code, not the moral worth of any tradition.

The dimensions are specified in advance rather than recovered from simulation. They are drawn from features that recur across documented enforcement regimes: whether conduct is observable enough to be judged, whether enforcing carries a reward, whether the authority to sanction is delegated to a designated few, and whether members can leave. The model does not identify which structural properties matter; it establishes what follows once these are taken as the governing properties. The division of labour is therefore explicit: the framework carries the theoretical commitment, and the model carries the deduction. A reader who doubts the dimensions is doubting the framework rather than the derivation, and the place to settle that doubt is the historical record rather than a further simulation.

Legibility (σ) refers to the degree to which compliance can be externally observed and socially verified. A system is highly legible when observers can classify adherence through visible acts (dress, attendance, dietary practice, ritual performance, verbal formulae) without privileged access

to internal states. A system is low in legibility when compliance depends on dispositions that cannot be reliably assessed by third parties. Legibility determines the feasibility of monitoring.

Substitutability captures whether externally observable compliance can function as a sufficient condition for moral standing. In high-substitutability systems, ritual performance or formal adherence terminates moral evaluation: an agent who fulfills the visible requirements is classified as moral regardless of internal state. In low-substitutability systems, external acts are explicitly necessary but insufficient; moral status depends on interior transformation that cannot be replaced by proxy performance.

Substitutability is analytically distinct from legibility. A system may render compliance visible (high legibility) without allowing visible compliance to substitute for interior reform (low substitutability). The joint condition (high legibility and high substitutability) is what makes proxy compliance a stable strategy: agents can achieve moral standing through externally performable acts alone, and observers accept this.

In the agent-based model, the parameter σ represents the joint legibility-substitutability axis. Formally, σ is the weight of external compliance (s_i) relative to internal cultivation (x_i) in the perceived moral score: $m_i = (1 - \sigma)x_i + \sigma s_i$. At low σ , moral evaluation depends on internal states that are opaque to observers; monitoring is infeasible and enforcement is unrewarding. At high σ , moral evaluation terminates at visible performance; monitoring is cheap and enforcement becomes structurally viable. The model collapses legibility and substitutability into a single parameter because in most codified systems these covary: systems that make compliance externally visible also tend to accept visible compliance as sufficient for moral standing. The theoretical framework distinguishes them because they can separate (monastic behavioral codes, for instance, render compliance visible without treating external observance as sufficient for spiritual attainment). The model tests the joint condition because it is the joint condition that creates the enforcement niche. Concrete indicators of high- σ systems include dress-code compliance, attendance records, dietary markers, public prayer performance, and verbal orthodoxy tests. Concrete indicators of low- σ systems include meditative attainment stages, sincerity of intention, character transformation, and contemplative depth. The

separate observability parameters (v_{obs} , a_{obs}) in the model handle the pure observation-accuracy dimension: v_{obs} captures how reliably external compliance signals are perceived, while a_{obs} captures the (typically very low) probability that internal states are observable.

Enforcement affordance refers to the ease with which deviations can be detected, classified, and sanctioned. Systems with binary violation categories (permitted/forbidden), well-defined infraction types, and established institutional sanction pathways possess high enforcement affordance. Systems characterized by interpretive ambiguity, graduated evaluation, and contested authority possess low enforcement affordance. High enforcement affordance reduces the per-act cost of policing.

Centralization of adjudication describes whether interpretive and disciplinary authority is monopolized or distributed. Under high centralization, a recognized clerical, judicial, or institutional body defines orthodoxy, controls appeals, and limits interpretive plurality. Under low centralization, multiple legitimate interpretive authorities coexist, norm enforcement is local and heterogeneous, and no single body can authoritatively define deviation. Centralization does not itself produce enforcement, but it amplifies enforcement once initiated by coordinating sanctioning across a population.

Exit cost refers to the social, economic, legal, and identity penalties associated with leaving or dissenting from the system. When exit cost is high (apostasy carries severe sanctions, kinship networks are internal to the community, economic livelihood depends on membership, identity foreclosure makes departure psychologically costly), disaffected members are trapped. Dissent is expressed internally rather than through departure. When exit cost is low, enforcement escalation produces fragmentation rather than compliance: members leave rather than submit. Exit cost therefore determines whether tightening leads to collapse or persistence.

2.3 From Geometry to Regimes

The interaction of these dimensions generates distinct structural regimes. Four are distinguished here; each has historical correlates, and each is realised in the simulations reported below.

In the *quiet regime*, legibility and enforcement affordance are low, or enforcement reward is

insufficient to sustain sanctioning as an adaptive strategy. Punishment remains negligible, exit is low, and the system persists without generating an enforcement niche. The code may be codified, but its structure does not render compliance auditable or sanctioning profitable.

In the *collapse regime*, legibility and enforcement affordance are high but exit cost is low. Enforcement escalation triggers departure. The population shrinks, monitoring loses targets, and the system fragments. Tightening is self-defeating.

In the *capture regime*, legibility, substitutability, and enforcement affordance are jointly high and exit cost is also high. Enforcement escalation does not produce mass departure. Sanctioning stabilizes because it is rewarded: enforcers gain status, institutional access, or material benefits. Compliance becomes self-reinforcing. The system converges on an enforcement-dominated equilibrium in which visible compliance is near-universal but may coexist with private dissent.

In the *mixed regime*, exit is feasible but non-trivial, and membership confers benefits sufficient to retain a substantial fraction of the population. Partial exit coexists with a hardened enforcement core. Overall prevalence may decline, but active participants exhibit higher rigidity and enforcement intensity than the pre-tightening population. The system persists in reduced form.

These regimes arise from structural incentives. They do not require abnormal psychological distributions, charismatic leaders, or exogenous political shocks, though all of these can shift parameters and change which regime the system occupies.

3 Methods

The model is documented below in ODD protocol order. Online Resource 1, Section S3, carries the extended model description: the full design-concept narrative, the recorded output fields, and the measurement justification.

3.1 Purpose and patterns

The model asks when a highly observable behavioral code can support an enforcement institution, and when restrictions on exit convert activated enforcement into durable capture. Three patterns evaluate it: whether an apparatus forms, measured as step-level punishment activation among active agents; whether it retains members, measured as the terminal exit rate; and whether sanctioning concentrates, measured as the top-share and the Gini of punishment counts. The model is adequate to the extent that these patterns respond to distinct structural inputs rather than moving together.

3.2 Entities, state variables and scales

The entities are individual agents embedded in a social network and one system-level institutional controller. The enforcer cadre is a collective defined over agents rather than a separate entity.

Each agent carries intrinsic cultivation x_i , belief position y_i , literalism L_i , rigidity b_i , enforcement propensity r_i , a binary externally legible compliance marker s_i , accumulated utility, institutional capital, accumulated service, an enforcer flag, an exit-intent counter, a disengagement flag, an exited flag, and a network position. Literalism is fixed; the remaining quantities change during a run. The controller carries a threat level T_t , an authority level A_t , a patronage budget, the current enforcement reward and punishment severity, the heresy-distance threshold, the orthodox belief position, and the current outside-option degradation δ_t .

Time is discrete and one step is the only temporal unit; the model makes no claim about the calendar duration a step represents. Space is not represented. Each reported run begins with $N = 350$ agents and proceeds for 450 steps, and each cell uses 30 independent seeds, numbered 1 through 30.

3.3 Parameter-observable mapping

Four correspondences connect model inputs to institutional features. The substitution weight σ represents how strongly public conduct, rather than opaque interior states, determines perceived moral

standing. Enforcement reward π , net of its cost κ , represents status, promotion, resources, or authority attached to successful sanctioning. The privilege bundle represents the standing advantages of an operative cadre, and exit conditions represent the relational, economic, legal, and cultural resources that make departure practicable. Online Resource 1, Sections S1 to S3, supplies the indicator table, parameter defaults and tested ranges, and the extended measurement justification.

3.4 Process overview and scheduling

At the start of a step the controller applies or relaxes threat and, with it, authority, the patronage budget, punishment reward and severity, and the heresy threshold; unless orthodoxy is fixed, it also updates the orthodox belief position. Agents are then shuffled and update their states. The model next recomputes mean literalism among active agents, clears and reselects the cadre, resets punishment counters, determines whether monopoly is active, and executes the shuffled punishment phase. It then computes active-agent punishment incidence, distributes patronage, and updates δ_t . Agents are finally reshuffled and attempt exit; successful exits remove edges, and the clock is incremented after exit. Metrics are recorded immediately before each ordered step, so the committed files contain rows indexed $t = 0, \dots, 449$.

Agent order is randomised afresh within each phase, and updates within a phase are asynchronous: an agent acts on the state left by agents that moved earlier.

3.5 Design concepts

Online Resource 1, Section S3, gives the full narrative for the concepts summarised here.

Basic principles. Legibility is a property of the code rather than of individuals, sanctioning is institutionally rewarded rather than privately costly, and the outside option governs whether tightening produces departure or compliance.

305 **Emergence.** Activation, the terminal exit rate, and the distribution of punishment are outcomes
306 of agent and controller rules; no agent begins with an enforcement role, and cadre membership is
307 recomputed each step.

308 **Adaptation.** Agents adapt without optimising: compliance display responds to local norms, cul-
309 tivation to a noisy investment decision, and exit intent to utility, punishment received, degree, and
310 the membership reward.

311 **Objectives.** No agent maximises an objective function; accumulated utility enters probabilistic
312 decision rules, principally the exit-desire logistic.

313 **Learning.** A favourable punishment outcome raises r_i by 0.04 and b_i by 0.03, an unfavourable
314 one lowers them by half those amounts, and both are clipped to $[0, 1]$. Literalism does not change.

315 **Prediction.** Agents make no forecast and decide on current local information alone.

316 **Sensing.** Neighbours are observed imperfectly: a compliance marker with probability $v_{\text{obs}} =$
317 0.90, cultivation with $a_{\text{obs}} = 0.05$, and a belief position, for the heresy-distance term, with $h_{\text{obs}} =$
318 0.15. The controller senses population-level quantities directly.

319 **Interaction.** Interaction is network-mediated in the compliance, cultivation, and punishment phases,
320 and global through the controller, which sets reward, severity, heresy threshold, and authority and
321 distributes patronage.

322 **Stochasticity.** One seeded generator drives network generation, initial traits, observation, compli-
323 ance and punishment probabilities, backlash, the exit draws, edge removal, and per-phase shuffling.

324 **Collectives.** The enforcer cadre is emergent and reselected every step with no persistent member-
325 ship; the controller is imposed and holds no agent state beyond the variables listed above.

326 **Observation.** Each run writes a step-level file of population quantities and an agent-level file of
 327 terminal agent records.

Outcome classification. The classifier treats activation and retention as separate axes: a run is active when `max_punish` is at least 0.10 and retained when `final_exit_rate` is at most 0.20, so

$$\text{CAPTURE} = \mathbf{1}\{\text{final_exit_rate} \leq 0.20\} \mathbf{1}\{\text{max_punish} \geq 0.10\}. \quad (1)$$

328 Inactive runs are QUIET, active and not-retained runs are MIXED, and active and retained runs
 329 are CAPTURE. COLLAPSE, defined descriptively by exit at least 0.90, is a tail within the inactive
 330 region: 100 percent of committed collapse runs have `max_punish` below 0.10. The joint robustness
 331 band for the activation threshold is $[0.075, 0.15]$, so continuous surfaces are the primary results and
 332 the four-class discretization a secondary summary.

Concentration and incidence metrics. Punishment incidence is computed only among active agents: `max_punish` is the maximum across recorded steps of the fraction of then-active agents punished at least once in that step, and `final_exit_rate` is the terminal exited fraction of the original population. For the privilege ablation, let c_i be cumulative punishments issued by agent i among the n_a agents active at the end of the run. Then

$$\text{top5pct_share_active} = \frac{\sum_{i \in \mathcal{K}} c_i}{\sum_{i=1}^{n_a} c_i}, \quad |\mathcal{K}| = \max\{1, \lceil 0.05n_a \rceil\}, \quad (2)$$

where $\mathcal{K}$ contains the largest counts, and with $c_{(i)}$ sorted ascending the active-agent Gini is

$$\text{gini_active} = \frac{n_a + 1 - 2 \sum_{i=1}^{n_a} \left(\sum_{j=1}^i c_{(j)} \right) / \sum_{j=1}^{n_a} c_{(j)}}{n_a}, \quad (3)$$

333 set to zero for an empty or zero-total vector. The random-allocation null holds the observed volume
 334 and n_a fixed, allocates events multinomially with equal probability $1/n_a$, repeats this 100 times us-
 335 ing the run seed, and records the median simulated Gini. `enforcer_punish_share` is descriptive

336 only: it saturates near 0.97 wherever punishment occurs.

337 **3.6 Initialisation**

338 Initialisation is driven by the run seed, so it varies across replicates but not across conditions
339 at a given seed. Intrinsic cultivation is $x_i \sim \text{clip}_{[0,1]} \{\mathcal{N}(0.35, 0.12^2)\}$, belief position is $y_i \sim$
340 $\text{clip}_{[0,1]} \{\mathcal{N}(0.50, 0.12^2)\}$, literalism is $L_i \sim \text{Beta}(1.5, 4.5)$, and rigidity and enforcement propen-
341 sity are drawn independently as $b_i, r_i \sim \text{Beta}(2, 4)$. The compliance marker begins at $s_i = 0$;
342 utility, capital, and service begin at zero; and all agents enter as non-exited members without an
343 enforcement role.

The reported sweeps use a Barabási–Albert scale-free network with attachment parameter

$$m = \max\{2, \text{round}[\log(N + 1)]\} = 6 \quad (4)$$

344 for $N = 350$, generated from the run seed. The controller begins with threat, and therefore author-
345 ity, at zero, the patronage budget at its constant term, and the closure level at the cell’s imposed
346 baseline δ_0 .

347 **3.7 Input data**

348 The model uses no external input data. Every driver is endogenous to the agent and controller rules
349 or fixed by a parameter for the run, including the threat schedule, which is a list of step indices
350 rather than an empirical series. No reported result is calibrated to observations. The only file read
351 at launch is the calibration parameter set committed at `results/BEST_PARAMS_frozen.json`,
352 whose contents are given in Online Resource 1.

3.8 Submodels

Perception and the code-geometry parameter. The code-geometry parameter $\sigma \in [0, 1]$ weights externally legible compliance against intrinsic cultivation in the perceived-morality signal,

$$\widehat{m}_{ij} = (1 - \sigma)\widehat{x}_{ij} + \sigma\widehat{s}_{ij}. \quad (5)$$

Failed compliance observation is coded as zero; failed cultivation observation is replaced by a draw from $\text{clip}_{[0,1]}\{\mathcal{N}(0.35, 0.25^2)\}$. The same geometry enters the legibility index

$$\ell(\sigma) = \sigma v_{\text{obs}} + (1 - \sigma)a_{\text{obs}}, \quad (6)$$

and π is the coefficient on this index, so with effective enforcement cost $\kappa_{i,t}^{\text{eff}}$ the punisher’s net institutional return is

$$\Delta u_{i,t}^{\text{pun}} = \pi_t \ell(\sigma) - \kappa_{i,t}^{\text{eff}}. \quad (7)$$

Agent update. An active agent chooses visible compliance probabilistically from local compliance and enforcement norms, updates cultivation by a noisy investment decision, and drifts weakly toward the current orthodoxy. Each active agent also receives a per-step membership reward

$$g_{i,t} = 0.0634 + 0.08 \ell(\sigma) + 0.1055 T_t, \quad (8)$$

354 which is added to utility and therefore offsets exit desire. Capital and service decay every step
 355 at rates 0.005 and 0.02. Exited agents are excluded from compliance and enforcement, and their
 356 capital and service continue to decay.

Exit. Exit has distinct desire and opportunity components, with `exit_opportunity_base` = 0.30 and `exit_threshold` = -1.0 in all reported experiments. With degree $d_{i,t}$, threat T_t , and

current outside-option degradation δ_t , the raw opportunity is

$$B_t = \begin{cases} 0.30, & \delta_t = 0, \\ 0.30(1 - \delta_t), & \delta_t > 0, \end{cases} \quad (9)$$

$$p_{i,t}^{\text{raw}} = \text{logit}^{-1}(B_t - 0.03d_{i,t} - 2.908T_t), \quad (10)$$

$$\tilde{p}_{i,t} = 0.02 + 0.98 \left(p_{i,t}^{\text{raw}} \right)^{5.98}. \quad (11)$$

357 In legacy and exogenous modes $p_{i,t}^{\text{opp}} = \tilde{p}_{i,t}(1 - \delta_t)$ when $\delta_t > 0$; in decoupled mode the outer
 358 factor is not applied. Exit desire is a logistic function of the threshold minus accumulated utility,
 359 punishment received, degree, and membership reward, must accumulate for twelve steps, and must
 360 then be matched by a successful opportunity draw. Effective exit cost is $1.00(1 + 0.03d_{i,t}) + 0.25s_i$.
 361 A successful exit removes $\text{round}(0.90d_i)$ of the agent's incident edges, so the network is not fixed.

Outside-option closure. Let δ_0 denote the imposed baseline and η the drift rate, clipped by the code to $[0, 0.5]$. Drift updates only when the step-level fraction of active agents punished is at least 0.08, and conditional on that gate

$$\delta_{t+1} = \text{clip}_{[0,1]} \{ \delta_t + \eta(\delta_t^* - \delta_t) \}. \quad (12)$$

Exit capacity is an independent structural input, since closure driven by the enforcement activity it is meant to explain would yield capture by construction. In that specification

$$\delta_t^* = \min\{\delta_{\max}, \delta_0 + \kappa q_t\}, \quad (13)$$

362 with q_t the active-agent punishment incidence, $\delta_{\max} = 0.85$, and κ varied by the experiment. In
 363 exogenous mode no drift occurs and $\delta_t = \delta_0$ throughout.

Cadre selection. Among active agents the target cadre size is $Q_t = \text{round}(0.08N_t^{\text{active}})$, with a minimum of one when the quota fraction is positive. Agents with capital at least 0.25 are eligible first and are ranked by

$$z_i = 0.6 \text{cap}_i + 0.3L_i + 0.1(r_i b_i). \quad (14)$$

364 Remaining positions are filled by the highest-ranked ineligible agents.

Privilege bundle. The bundle combines exclusivity, coercive advantage, reduced cost and risk, and material reinforcement. When controller authority $A_t \geq 0.35$, the punishment monopoly excludes all non-enforcers from policing. Enforcers carry a punishment-probability multiplier of 1.5 whether or not monopoly is active; with monopoly off, non-enforcers receive the operative multiplier $\text{non_enforcer_punish_eps} = 0.02$, while the separately declared $\text{non_enforcer_punish_mult} = 0.25$ is passed by the sweeps but unused in this branch. Capital first reduces the base cost according to

$$\kappa'_{i,t} = 0.08 \left(1 - 0.20 \text{cap}_{i,t}\right), \quad (15)$$

365 and an enforcer pays $0.30\kappa'_{i,t}$, a further 70 percent role discount. Enforcer backlash probability is
 366 multiplied by 0.25. Every successful punishment adds 0.15 capital, capped at 2.0, and one unit of
 367 service. The nonnegative budget $0.15 + 0.60T_t$ is distributed only among active enforcers, equally
 368 if their total service is zero and otherwise in proportion to nonnegative accumulated service.

369 **Threat controller.** At scheduled shock times $t \in \{100, 220, 320\}$ threat increases by 0.10; other-
 370 wise it decays by a factor $1 - 0.03$. Authority is $A_t = 1 - \exp(-2T_t)$, the budget is $0.15 + 0.60T_t$,
 371 punishment reward and severity each rise by $0.20T_t$ from baseline, and the heresy-distance thresh-
 372 old falls by $0.08T_t$, subject to clipping. The shock-off factorial and all decoupled runs use an empty
 373 schedule, so threat is identically zero and every threat-gated term vanishes.

3.9 Experimental design

The shock factorial is committed at `recon/factorial_shock_off/sweep_seed_results.csv` and `recon/factorial_shock_on/sweep_seed_results.csv`. Each condition contains 4,050 runs from $\eta \in \{0, 0.05, 0.10, 0.20, 0.30\}$, $\delta_0 \in \{0.10, 0.20, 0.30\}$, $\sigma \in \{0.25, 0.75, 0.95\}$, $\pi \in \{0.05, 0.25, 0.50\}$, and 30 seeds. The off condition has no shocks; the on condition uses the schedule above.

The exogenous- δ dose-response is committed at `recon/exogenous_delta_fixed/sweep_seed_results.csv`. Its 1,620 runs cross $\delta_0 \in \{0, 0.20, 0.40, 0.60, 0.80, 0.95\}$, $\sigma \in \{0.25, 0.75, 0.95\}$, $\pi \in \{0.05, 0.25, 0.50\}$, and 30 seeds, with exogenous mode and $\eta = 0$.

The activation-boundary experiments are committed at `recon/boundary_open/sweep_seed_results.csv` and `recon/boundary_sealed/sweep_seed_results.csv`. Each contains 1,890 runs crossing $\sigma \in \{0.05, 0.15, 0.25, 0.35, 0.45, 0.55, 0.65, 0.75, 0.95\}$, $\pi \in \{0.01, 0.03, 0.05, 0.10, 0.15, 0.25, 0.50\}$, and 30 seeds. Both use exogenous mode and $\eta = 0$; the open boundary fixes $\delta_0 = 0$, and the sealed boundary fixes $\delta_0 = 0.95$.

The privilege ablation is committed at `recon/privilege_ablation/ablation_seed_results.csv`. The floor and ceiling each contain 120 runs crossing $(\sigma, \pi) \in \{(0.75, 0.25), (0.75, 0.50), (0.95, 0.25), (0.95, 0.50)\}$ with 30 seeds, both at $\delta_0 = 0.95$ in exogenous mode with no drift and no shocks. The floor disables monopoly and the cadre quota, equalizes punishment multipliers at 1.0, removes capital gain, capital and role cost discounts, and backlash protection, and sets the patronage budget to zero. The ceiling retains the complete default bundle described above.

3.10 Reproducibility

The implementation is written in Python using Mesa 3.4.2 (Kazil et al. 2020), NumPy 2.4.2 (Harris et al. 2020), pandas 3.0.0 (McKinney 2010), and NetworkX 3.6.1 (Hagberg et al. 2008), with versions pinned in `requirements.txt`. The seven parameters that the sweep driver reads at launch from `results/BEST_PARAMS_frozen.json` are listed in Online Resource 1, Section S2, and the values stated above are those values. They were confirmed by replaying single runs from three

sweeps against the frozen model source and reproducing the committed `metrics.csv` and `agent_summary.csv` exactly.

The canonical classifier in `src/regime_classifier.py` is the single source of truth for discrete outcomes. All reported regimes are computed from continuous `final_exit_rate` and `max_punish`; stored regime columns are non-authoritative.

4 Results

4.1 Observability and reward jointly activate enforcement

Enforcement activates along a joint frontier in observability and reward: neither input activates the apparatus by itself. Table 1 reports 1,890 open-exit runs, with 30 runs in each of $63 \sigma \times \pi$ cells. The frontier slopes downward from the upper-right active region toward the lower-left inactive region. At $\sigma = 0.05$, activation first appears at $\pi = 0.25$, where 0.17 of runs activate, and reaches 0.90 at $\pi = 0.50$. At $\sigma = 0.25$, the corresponding transition occurs earlier: activation is 0 through $\pi = 0.05$, 0.63 at $\pi = 0.10$, and 0.90 at $\pi = 0.15$. At $\sigma \geq 0.75$, every reward level at or above 0.03 activates in all 30 runs per cell. This diagonal pattern is a frontier rather than a threshold on either axis.

The low-reward column remains inactive at every observability level. For $\pi = 0.01$, activation is 0 in all nine cells, or 0 of 270 runs, even as σ rises from 0.05 to 0.95. High observability makes deviations classifiable, but a classifiable deviation does not support an apparatus when sanctioning supplies no adequate return. Conversely, high reward cannot activate policing where conduct remains too opaque to classify: at $\sigma = 0.05$, the five columns through $\pi = 0.15$ remain at 0 in 150 runs. The joint threshold therefore has a substantive interpretation: observability without payoff yields no apparatus, while payoff without observability yields nothing reliably sanctionable. Across the full open-exit grid, 0.598 of 1,890 runs activate, and mean maximum punishment ranges from 0.033 to 0.181. The claim applies to the tested observability and reward grid, the model’s privilege architecture, and the canonical activation threshold of 0.10.

Table 1: Probability of activation with exit open ($\delta_0 = 0$); 30 runs per cell.

$\sigma \backslash \pi$	0.01	0.03	0.05	0.10	0.15	0.25	0.50
0.05	0.0	0.00	0.00	0.00	0.00	0.17	0.90
0.15	0.0	0.00	0.00	0.00	0.40	0.90	1.00
0.25	0.0	0.00	0.00	0.63	0.90	1.00	1.00
0.35	0.0	0.00	0.00	0.93	1.00	1.00	1.00
0.45	0.0	0.00	0.60	1.00	1.00	1.00	1.00
0.55	0.0	0.00	1.00	1.00	1.00	1.00	1.00
0.65	0.0	0.27	1.00	1.00	1.00	1.00	1.00
0.75	0.0	1.00	1.00	1.00	1.00	1.00	1.00
0.95	0.0	1.00	1.00	1.00	1.00	1.00	1.00

4.2 Institutional design concentrates punishment

Institutional privilege, rather than agent heterogeneity alone, produces most of the observed concentration of punishment. The ablation compares 120 floor runs without the privilege bundle with 120 ceiling runs containing the full bundle. Removing the bundle eliminates the designated cadre and its asymmetric access to sanctioning opportunities. The contrast therefore isolates the contribution of institutional allocation after leaving the same underlying agent population and interaction rules in place.

At the floor, the top five percent of active agents issue 0.091 of punishments and the active-agent Gini is 0.240. At the ceiling, the corresponding values rise to 0.580 and 0.901 (Table 2; Figure 4). The floor is not perfectly equal: its Gini of 0.240 exceeds the random-allocation null of 0.028. That residual is the concentration generated by stochastic encounters and heterogeneous opportunities even without assigned privilege. It is small relative to the ceiling and shows why the null matters: zero privilege does not imply mechanically uniform punishment. The floor-to-ceiling difference accounts for 0.843, or 84.3 percent, of the ceiling top-five-percent share. This result applies to the floor and ceiling architectures tested over 120 runs per arm.

Table 2: Concentration and exit under the privilege floor and ceiling; 120 runs per arm.

Arm	Top5 active	Gini active	Random-null Gini	Exit rate
Floor	0.091	0.240	0.028	0.585
Ceiling	0.580	0.901	0.062	0.331

The add-back experiment clarifies the architecture. Seven of eight privileges reproduce the floor values to four decimal places when added alone. The punishment multiplier is the exception: across 120 runs it raises the top-five-percent share to 0.2505 and the Gini to 0.6317, recovering 32.7 percent of the floor-to-ceiling span. It operates on a mean of 393 punishments per run, compared with 75,612 at the floor and 19,101 at the ceiling, so this isolated recovery describes the distribution of a small event count.

Concentration also fails to identify whether an active system is captured. Across 320 captured and 1,120 merely active runs, the top-five-percent shares are 0.678 and 0.625, and Gini values are 0.918 and 0.903. Exit rates, by contrast, are 0.105 and 0.339. Empirically, a concentrated enforcement record can reveal an active apparatus, but it cannot distinguish an apparatus that retains its subjects from one that remains leaky. Enforcement regimes must therefore be identified with retention evidence, not with sanction concentration alone. This comparison is conditional on systems crossing the activation criterion; quiet systems are not pooled with active ones.

4.3 Exit foreclosure retains an active apparatus

Exit foreclosure converts active systems into captured systems in a steep, monotonic dose-response, but it cannot activate an otherwise inactive cell. The imposed-closure experiment contains 1,620 runs: 30 runs in each of $54 \sigma \times \pi \times \delta_0$ cells. From $\delta_0 = 0$ through 0.4, capture remains 0 in each 270-run closure level. At 0.6, capture first appears at 0.011, or 3 of 270 runs. It then rises sharply to 0.285 at 0.8, or 77 of 270 runs, and to 0.889 at 0.95, or 240 of 270 runs. The largest change is concentrated between imposed closure 0.6 and 0.95: a marginal captured tail becomes nearly the entire tested grid's active region. Pooled mean exit falls over the same sequence from 0.332 at $\delta_0 = 0$ to 0.288 at 0.6, 0.216 at 0.8, and 0.072 at 0.95. The figure therefore shows a capacity transition, not a linear response: moderate degradation leaves enough outside opportunity for departure, while near-complete degradation retains most members exposed to an active apparatus.

The inactive cell supplies the scope condition. At $\sigma = 0.25, \pi = 0.05$, capture remains 0 at all six closure levels: 0 of 180 runs overall and 0 of 30 even at $\delta_0 = 0.95$. Mean maximum

Table 3: Capture rate by imposed exit closure; 270 runs per closure level.

δ_0	0.0	0.2	0.4	0.6	0.8	0.95
Capture rate	0.000	0.000	0.000	0.011	0.285	0.889

punishment stays at 0.051 as mean exit falls from 0.044 to 0.002. The zero is substantively important because closure successfully prevents departure in this cell but never creates sanctioning activity. An inactive cell at every closure level shows that retention is not itself enforcement. The result applies to an imposed outside-option degradation; it does not claim that every empirical restriction on exit follows the same functional form.

4.4 Two sources of exit foreclosure

Endogenous exit closure occurs when institutional absorption outruns the renewal of members' outside ties and economic independence. In words, outside connections disappear faster than contact can replenish them, so the stock that makes departure practicable contracts toward its floor. The closure sweep contains 15 runs, five at each tenure crowd-out level. Mean final capacity is 0.9302 at crowd-out 0.0005, 0.8392 at 0.002, and 0.5166 at 0.008. At the highest value, tie absorption exceeds base renewal and closes the tie channel; aggregate capacity remains above zero because the economic channel stays open in this experiment. This is a capacity result, not a claim that every agent who has capacity exits.

The generational channel changes the composition of the membership stock. Born-inside cohorts inherit fewer outside ties and less independent economic position than recruits who arrive with resources formed elsewhere. With the economic channel active, born-inside replacement lowers terminal aggregate exit capacity in all 40 seed-matched comparisons, ten runs in each of four parameter cells. The reduction is approximately 0.10 to 0.15 across those cells. Realized cumulative exit is lower in all four cell means but in 32 of 40 individual seed comparisons. Capacity therefore moves monotonically in these comparisons, while realized departure also depends on whether agents develop and sustain exit intent. Distinguishing the two prevents a lower observed exit count from being treated as a direct measure of diminished opportunity. The capacity update

uses ties, economic independence, tenure, and generic membership benefit, with no punishment or enforcer input. The claim is limited to the tested renewal, absorption, dependence, and born-inside settings over five or ten runs per cell.

4.5 Robustness

The activation and closure findings hold jointly across activation thresholds from 0.075 to 0.15, with 0.10 used in the tables. Threat shocks leave the 4,050-run regime distribution unchanged: 450 quiet, 720 mixed, and 2,880 captured with or without shocks. All primary runs use one Barabási–Albert topology, one population size, and the reported parameter grids.

5 Discussion

5.1 Implications

The results answer the opening puzzle by separating cohesion from control. Strict requirements can select committed entrants and coordinate visible conduct, but those functions do not establish why members remain when they wish to leave. The activation frontier shows how observable rules and rewards create an enforcement apparatus. The closure experiments show that the apparatus retains members only when their outside option deteriorates. Durability can therefore arise from commitment, foreclosure, or both, and the same observed persistence does not identify which mechanism operates.

This separation changes how enforcement regimes are explained. Activation is an organizational supply problem: conduct must be legible and sanctioning must return enough status or resources to sustain action. Concentration is an allocation problem: privileges assign sanctioning opportunities to a cadre. Capture is a retention problem: the outside resources needed for departure must be unavailable or eroding. Treating all three as “strictness” collapses distinct causes into one label.

The concentration result is especially consequential for empirical inference. A highly skewed

punishment record does not show that the regime has captured its members. Captured and merely active systems display nearly the same concentration while their exit rates differ threefold. Researchers must observe outside options and realized departure alongside disciplinary records. Likewise, the floor ablation shows that some concentration arises without privilege, but most of the ceiling pattern is institutionally manufactured. The relevant explanation is the distribution of authority and opportunity, not a presumed concentration of punitive dispositions.

The two axes remain coupled at the activation margin. Sealing exit raises activation in 12 of 63 frontier cells while leaving 51 unchanged. Retention is therefore analytically distinct from activation without being universally orthogonal to it. Longer exposure supplies additional targets and rewards that can push marginal systems across the activation frontier. That coupling reinforces the need to measure both processes directly.

Imposed closure can convert activated regions to capture, but cannot manufacture capture where enforcement lacks both legibility and reward. The dose-response therefore establishes that exit closure is necessary but not sufficient within the tested design. Exit capacity is an independent structural condition in the main intervention, while the internal-absorption extension updates capacity without enforcement inputs.

These results are entailments of a specification rather than discoveries about which dimensions govern enforcement. The dimensions are fixed before any run, so nothing in the runs can vindicate the choice of them. Whether they are the right dimensions is settled by the historical record that motivated them and by the predictions they generate, not by further simulation. The predictions set out below are therefore addressed to cases that played no part in specifying the model.

5.2 Enforcement without intention

The account is structural in the strict sense. No agent in it holds a doctrine, believes anything, or intends domination. Agents occupy positions defined by four conditions: how visible their conduct is, what enforcing returns to them, whether the authority to sanction is delegated to a designated few, and whether an outside option remains available. Enforcement concentrates in a small cadre

and populations lock in without any actor preferring either outcome. The pattern is a property of the arrangement of positions, not of the dispositions of those who occupy them.

This cuts against the intentionalist account, which explains such regimes by the character of the people who staff them: zealots, true believers, opportunists. Zeal is not required. A position that combines visibility, institutional return, delegated authority, and foreclosed exit produces concentrated sanctioning whoever fills it. What a content-based explanation cannot easily accommodate is the recurrence of similar enforcement forms across traditions that share no doctrinal content; a positional account predicts that recurrence, because the same positions can be assembled from any doctrine. Conviction and enthusiasm are present in historical regimes and shape their texture, but they are not what generates the form.

5.3 Historical scope

The mechanism is sufficient to generate a regularity documented independently by historical scholarship: enforcement-dominant regimes co-occur with foreclosed exit, whereas demanding communities with open exit can sustain enforcement without durable lock-in. Inquisitorial enforcement combined doctrinal surveillance with state capacity to prevent flight (Given 1997). Residential and boarding schools severed family and language ties that otherwise supported an outside option (Adams 1995). Internally reproducing communities provide a contrasting basis for evaluating persistence under demanding doctrine without equating doctrine with coercion (Hostetler 1974). Historical outcomes combine multiple mechanisms. This account isolates one sufficient mechanism among them. Its unit is the bounded regime, not a religion across places and centuries.

5.4 Predictions

First, high observability produces an apparatus only where enforcement carries a positive institutional return. The warrant is the activation frontier: the $\pi = 0.01$ column remains inactive in 0 of 270 activations across every observability level, while low observability also blocks all but the highest rewards. The prediction is falsified if highly observable regimes with no status, mate-

rial, or authority return to sanctioning repeatedly develop stable, specialized enforcement at rates comparable to rewarded regimes.

Second, codified communities with open exit exhibit active but leaky enforcement rather than capture. The warrant is that 0 of 1,890 open-exit runs reach capture despite an active rate of 0.598. The prediction is falsified if independently measured open outside options coexist with durable retention under active enforcement, without selection, replacement, or another mechanism accounting for low departure.

Third, punishment concentration does not distinguish captured regimes from regimes that are merely active. The warrant is the close correspondence in top-five-percent shares, 0.678 versus 0.625, and Gini values, 0.918 versus 0.903, across 320 captured and 1,120 active runs, while exit rates are 0.105 versus 0.339. The prediction is falsified if concentration consistently separates captured from active systems after activation intensity and sanctioning opportunity are held constant.

Fourth, internally reproducing communities lose exit capacity across generations without requiring an increase in coercion. The warrant is that born-inside replacement lowers capacity in all 40 seed-matched comparisons while the update contains no enforcement input. The prediction is falsified if born-inside cohorts retain outside ties and economic independence at the same levels as matched recruits, or if capacity declines only when punishment enters the process.

These tests require measures fixed independently of the outcome. The scoring unit is a historically bounded regime durable enough for stable enforcement to be meaningful, not a whole tradition or brief mobilization. Observability, enforcement return, operative privilege, and exit capacity must be recorded separately. The design must include demanding systems with open exit and constrained-exit systems with weak activation conditions. Systematic false positives or false negatives revise the theory rather than relabeling the cases.

The framework also speaks to contemporary settings, though what it speaks to is the durability of enforcement rather than the mobilisation that produces it. Where exit capacity is high, so that departure carries no legal penalty, livelihoods do not run through the community, and members retain dense ties beyond it, the account predicts active but leaky enforcement rather than durable

control. Enforcement intensity should therefore not translate into retention: rising enforcement activity should be observed alongside continuing disaffiliation, and the two should be measured separately rather than read off one another. Lock-in, where it occurs, should be found not in the movements that enforce most visibly but in subcommunities where exit is locally foreclosed. The indicators of that foreclosure are relational and material rather than doctrinal: outside ties severed rather than merely thinned; employment, housing, and schooling routed through the community; children raised without sustained relationships beyond it. Retention should track exit foreclosure rather than enforcement intensity, so a design that samples on visible enforcement will mistake activity for durability. The informative negative cases are demanding communities under open exit, where requirements are heavy and departure remains practicable, and any serious test must include them. These are applications of the framework rather than results, since no contemporary case has been scored or simulated here.

5.5 Limitations

The findings apply to the reported behavioral rules and parameter grids. The primary simulations use a Barabási–Albert network, 350 agents, and a 450-step horizon. The model does not establish invariance across topologies, population sizes, or longer horizons. The discrete classes summarize continuous enforcement and exit measures. Their activation threshold is robust over the tested band from 0.075 to 0.15.

Activation and retention are separable in 51 of 63 frontier cells. They are coupled at the margin in the remaining 12 cells. The internal-capacity module represents relational, economic, tenure, and generational processes. Legal, political, and cultural closure enter as external conditions. The module makes capacity independent of enforcement inputs by construction. It does not claim that enforcement can never alter capacity in historical systems.

Capacity and realized exit are different outcomes. Having an outside option does not imply that an agent chooses departure. Observed departure also reflects exit intent and the simulation horizon. The model represents one enforcer role and one institutional controller. It does not represent

competing jurisdictions, reform movements, or state intervention among multiple organizations.

Psychological predispositions and doctrinal truth are outside the causal mechanism. The results do not rank traditions or infer latent sincerity. Historical applications require regime-level measurement. The operational classes organize outcomes by activation and retention. They are not claims about universal long-run attractors.

The resulting framework is narrower and more testable. Empirical applications should measure observability, enforcement reward, operative privilege, and exit capacity independently of outcomes, include negative cases, and distinguish active but leaky enforcement from durable retention. The model traces what follows from the specified structural combinations across the explored parameter space. It does not establish an endogenous theory of capture or a general tendency of codified belief toward coercion.

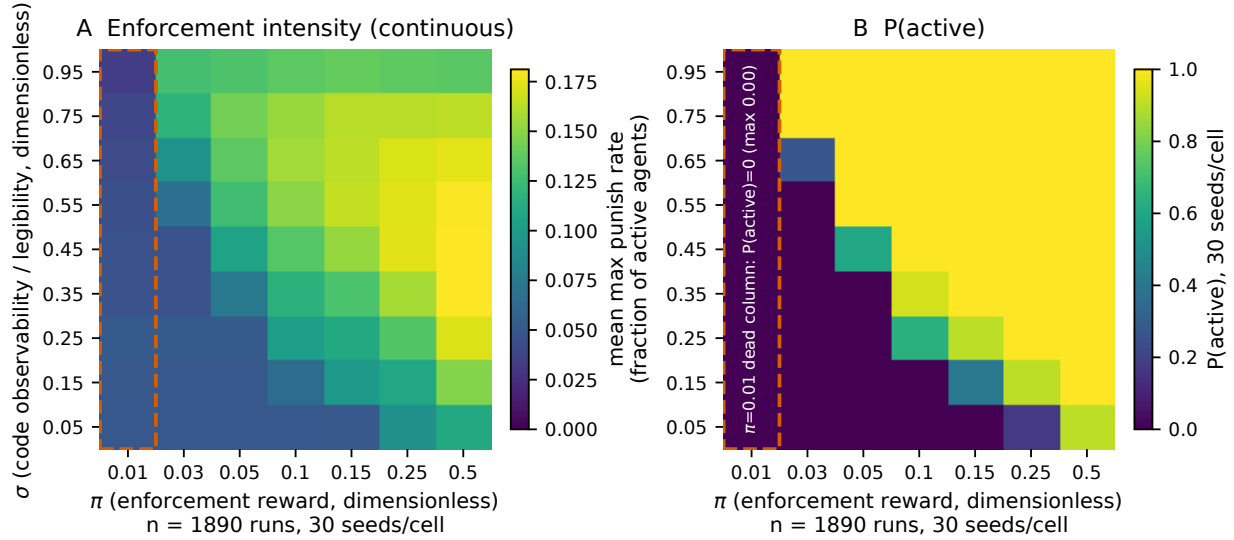


Fig 1: Activation frontier under open exit. Agent-based sweep at $\delta_0 = 0$ (exit open), $n = 1890$ runs ($9 \sigma \times 7 \pi$ cells $\times$ 30 seeds per cell). (A) Mean maximum punish rate, the continuous enforcement-intensity surface. (B) Probability that a run is classified active, recomputed with the hierarchical classifier (active: max punish rate ≥ 0.1); overall $P(\text{active}) = 0.598$. In the reported grid, enforcement activates only above a joint σ - π threshold: the $\pi = 0.01$ column (dashed outline) is inactive in all 9 cells, so no run in that column sustains an enforcement apparatus at any tested level of code observability.

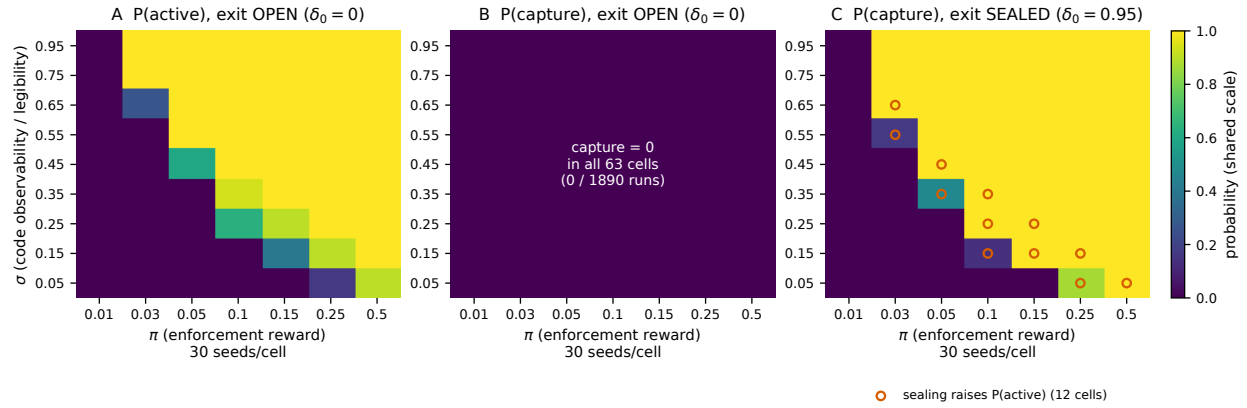


Fig 2: Exit closure is necessary for capture. All panels share one colour scale; $9 \sigma \times 7 \pi$ cells $\times$ 30 seeds per cell. (A) Probability a run is active with the exit open ($\delta_0 = 0$, $n = 1890$ runs): code geometry alone sets activation, overall $P(\text{active}) = 0.598$. (B) Probability of capture in those same 1890 runs: **0 of 1890 runs reach capture with the exit open**; no run with the exit open reaches capture, in any of the 63 cells, despite that active rate of 0.598. (C) Probability of capture with the exit sealed ($\delta_0 = 0.95$, $n = 1890$ runs): the active region in these runs now captures. Panels B and C are the same quantity under the two exit conditions, so the contrast isolates the effect of closing the exit. The two dimensions are not fully orthogonal: circled cells in (C) are the 12 of 63 cells where sealing also raises $P(\text{active})$ (mean shift over all cells $+0.062$; $+0.328$ among circled cells), and activation falls in no reported cell. Regimes recomputed with the hierarchical classifier.

Acknowledgments

References

- Adams DW (1995) Education for Extinction: American Indians and the Boarding School Experience, 1875–1928. University Press of Kansas, Lawrence, KS.
- Axelrod R (1986) An evolutionary approach to norms. *American Political Science Review* 80(4):1095–1111.
- Berman E (2000) Sect, subsidy, and sacrifice: An economist’s view of ultra-Orthodox Jews. *Quarterly Journal of Economics* 115(3):905–953.
- Berman E, Laitin DD (2008) Religion, Terrorism and Public Goods: Testing the Club Model. *Journal of Public Economics* 92(10–11):1942–1967.

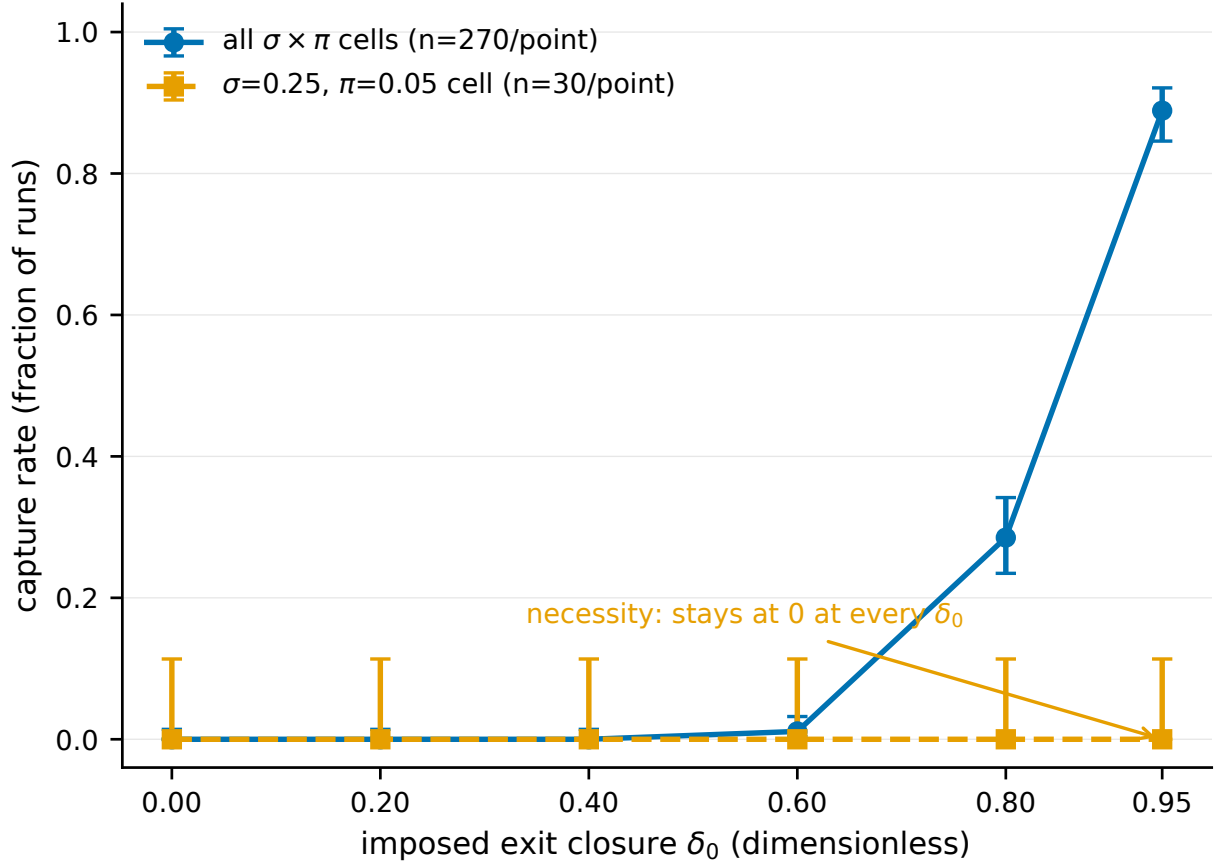


Fig 3: Exit-closure dose-response and the necessity of code geometry. Capture rate against imposed exit closure δ_0 ($n = 1620$ runs total; 270 runs per pooled point, 30 per cell-specific point), with Wilson 95% intervals. Pooled over all $\sigma \times \pi$ cells (circles) capture rises monotonically from 0.000 at $\delta_0 = 0$ to 0.889 at $\delta_0 = 0.95$. In the reported runs for the $\sigma = 0.25, \pi = 0.05$ cell (squares, dashed), the capture rate is 0.000 at every tested level, including $\delta_0 = 0.95$: imposed closure does not produce capture in this inactive cell. Regimes recomputed with the hierarchical classifier.

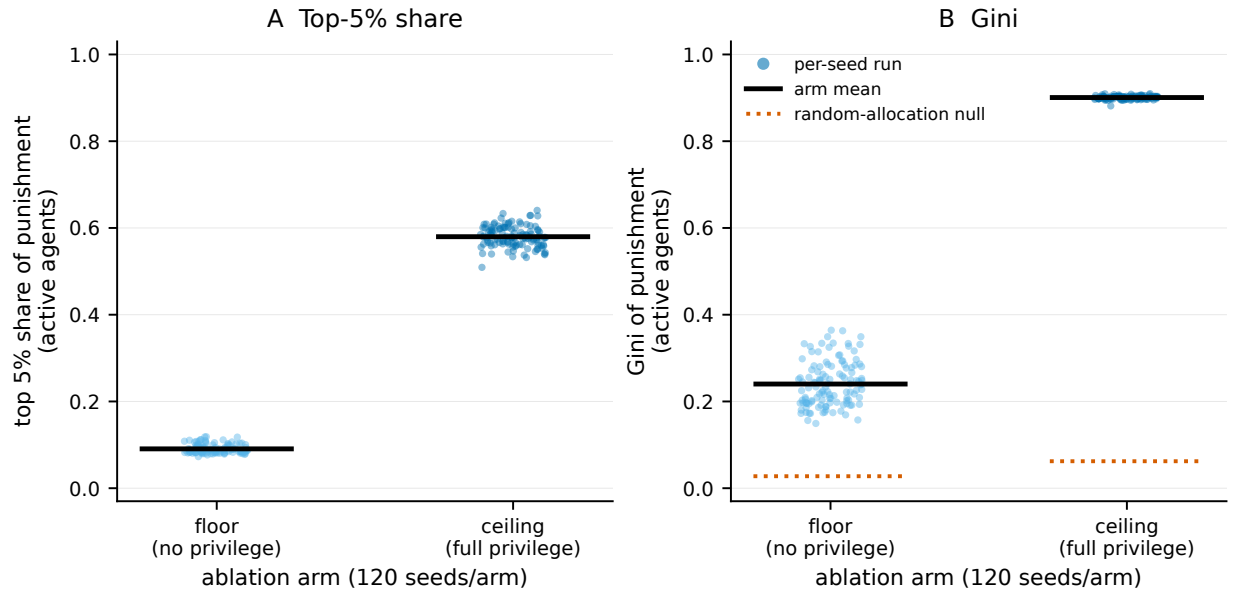


Fig 4: Concentration is manufactured by the privilege architecture. Role-independent concentration for the ablation floor (no privilege) and ceiling (full privilege bundle), 120 seeds per arm ($n = 240$ runs); points are individual runs, black bars arm means. (A) Top-5% share of punishment among active agents rises from 0.091 at the floor to 0.580 at the ceiling, so 84% of ceiling concentration in these runs is privilege-manufactured. (B) Gini rises from 0.240 to 0.901 against random-allocation nulls (dotted) of 0.028 and 0.062; the reported floor sits close to its null, so these runs do not show substantial concentration emerging from the unprivileged model. A committed random-allocation null exists only for the Gini statistic, so panel A carries no null reference.

- 637 Bok J (2025) Religious Exit Costs. *Journal for the Scientific Study of Religion* 64(2):227–239.
638 doi:10.1111/jssr.12963.
- 639 Boyd R, Gintis H, Bowles S (2010) Coordinated Punishment of Defectors Sustains Cooperation
640 and Can Proliferate When Rare. *Science* 328(5978):617–620. doi:10.1126/science.1183665.
- 641 Boyd R, Gintis H, Bowles S, Richerson PJ (2003) The evolution of altruistic punishment. *Proceed-*
642 *ings of the National Academy of Sciences* 100(6):3531–3535.
- 643 Bromley DG (1998) The Politics of Religious Apostasy: The Role of Apostates in the Transforma-
644 tion of Religious Movements. Praeger.
- 645 Carvalho JP (2013) Veiling. *Quarterly Journal of Economics* 128(1):337–370.
- 646 Carvalho JP (2019) Religious Clubs: The Strategic Role of Religious Identity. In: Carvalho JP, Iyer
647 S, Rubin J (eds) *Advances in the Economics of Religion*, International Economic Association
648 Series, vol 158. Palgrave Macmillan, Cham, pp 25–41.
- 649 Centola D, Willer R, Macy M (2005) The emperor’s dilemma: A computational model of self-
650 enforcing norms. *American Journal of Sociology* 110(4):1009–1040. doi:10.1086/427321.
- 651 Cook M (2000) Commanding Right and Forbidding Wrong in Islamic Thought. Cambridge Uni-
652 versity Press.
- 653 Cragun RT, Hammer JH (2011) One person’s apostate is another person’s convert: What termi-
654 nology tells us about pro-religious hegemony in the sociology of religion. *Humanity & Society*
655 35(1–2):149–175.
- 656 Engelstein L (1999) Castration and the Heavenly Kingdom: A Russian Folktale. Cornell University
657 Press, Ithaca, NY.
- 658 Fehr E, Gächter S (2002) Altruistic punishment in humans. *Nature* 415(6868):137–140.

- 659 Fehr E, Gächter S (2000) Cooperation and Punishment in Public Goods Experiments. *American*
660 *Economic Review* 90(4):980–994. doi:10.1257/aer.90.4.980.
- 661 Gelfand MJ, et al (2011) Differences between tight and loose cultures: A 33-nation study. *Science*
662 332(6033):1100–1104.
- 663 Given JB (1997) *Inquisition and Medieval Society: Power, Discipline, and Resistance in Languedoc*.
664 Cornell University Press.
- 665 Gore R, Lemos C, Shults FL, Wildman WJ (2018) Forecasting Changes in Religiosity and Existential
666 Security with an Agent-Based Model. *Journal of Artificial Societies and Social Simulation*
667 21(1):4. doi:10.18564/jasss.3596.
- 668 Greif A (2006) *Institutions and the Path to the Modern Economy: Lessons from Medieval Trade*.
669 Cambridge University Press.
- 670 Greif A, Mokyr J (2016) Cognitive rules, institutions, and economic growth: Douglass North and
671 beyond. *Journal of Institutional Economics* 13(1):25–52.
- 672 Harris CR, Millman KJ, van der Walt SJ, et al (2020) Array programming with NumPy. *Nature*
673 585(7825):357–362. doi:10.1038/s41586-020-2649-2.
- 674 Hagberg AA, Schult DA, Swart PJ (2008) Exploring Network Structure, Dynamics, and Function
675 using NetworkX. In: Varoquaux G, Vaught T, Millman J (eds) *Proceedings of the 7th Python in*
676 *Science Conference*, Pasadena, CA, pp 11–15.
- 677 Hauert C, Traulsen A, Brandt H, Nowak MA, Sigmund K (2007) Via Freedom to Coercion: The
678 Emergence of Costly Punishment. *Science* 316(5833):1905–1907.
- 679 Hilbe C, Traulsen A, Röhl T, Milinski M (2014) Democratic decisions establish stable authorities
680 that overcome the paradox of second-order punishment. *Proceedings of the National Academy*
681 *of Sciences* 111(2):752–756.

- 682 Hirschman AO (1970) Exit, Voice, and Loyalty: Responses to Decline in Firms, Organizations, and
683 States. Harvard University Press.
- 684 Hostetler JA (1974) Hutterite Society. Johns Hopkins University Press, Baltimore, MD.
- 685 Iannaccone LR (1988) A Formal Model of Church and Sect. *American Journal of Sociology*
686 94(Supplement):S241–S268. doi:10.1086/228948.
- 687 Iannaccone LR (1992) Sacrifice and Stigma: Reducing Free-Riding in Cults, Communes, and Other
688 Collectives. *Journal of Political Economy* 100(2):271–291.
- 689 Iannaccone LR (1994) Why Strict Churches Are Strong. *American Journal of Sociology*
690 99(5):1180–1211.
- 691 Iannaccone LR, Makowsky MD (2007) Accidental Atheists? Agent-Based Explanations for the
692 Persistence of Religious Regionalism. *Journal for the Scientific Study of Religion* 46(1):1–16.
- 693 Iyer S (2016) The New Economics of Religion. *Journal of Economic Literature* 54(2):395–441.
694 doi:10.1257/jel.54.2.395.
- 695 Kazil J, Masad D, Crooks A (2020) Utilizing Python for Agent-Based Modeling: The Mesa Frame-
696 work. In: Thomson R, Bisgin H, Dancy C, Hyder A, Hussain M (eds) *Social, Cultural, and Be-*
697 *havioral Modeling*, Lecture Notes in Computer Science, vol 12268. Springer, Cham, pp 308–317.
698 doi:10.1007/978-3-030-61255-9_30.
- 699 Kelly HA (1989) Inquisition and the prosecution of heresy: Misconceptions and abuses. *Church*
700 *History* 58(4):439–451.
- 701 Kuran T (1995) Private Truths, Public Lies: The Social Consequences of Preference Falsification.
702 Harvard University Press.
- 703 Mann M (1986) The Sources of Social Power, Vol. 1. Cambridge University Press.

- McBride M (2007) Club Mormon: Free-Riders, Monitoring, and Exclusion in the LDS Church. *Rationality and Society* 19(4):395–424. doi:10.1177/1043463107083736.
- McKinney W (2010) Data Structures for Statistical Computing in Python. In: van der Walt S, Millman J (eds) *Proceedings of the 9th Python in Science Conference*, pp 56–61. doi:10.25080/Majora-92bf1922-00a.
- Norenzayan A, Gervais WM (2013) The origins of religious disbelief. *Trends in Cognitive Sciences* 17(1):20–25.
- North DC (1990) *Institutions, Institutional Change and Economic Performance*. Cambridge University Press.
- Pierson P (2000) Increasing returns, path dependence, and the study of politics. *American Political Science Review* 94(2):251–267.
- Rubin J (2017) *Rulers, Religion, and Riches: Why the West Got Rich and the Middle East Did Not*. Cambridge University Press.
- Scott JC (1998) *Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed*. Yale University Press.
- Sigmund K, De Silva H, Traulsen A, Hauert C (2010) Social learning promotes institutions for governing the commons. *Nature* 466(7308):861–863.
- Streib H, Hood RW, Keller B, Csöff RM, Silver CF (2009) Deconversion: Qualitative and Quantitative Results from Cross-Cultural Research in Germany and the United States of America. Vandenhoeck & Ruprecht.
- Tilly C (1985) War making and state making as organized crime. In: Evans P, Rueschemeyer D, Skocpol T (eds) *Bringing the State Back In*. Cambridge University Press, pp 169–191.
- Weber M (1946) Politics as a Vocation. In: Gerth HH, Mills CW (eds) *From Max Weber: Essays in Sociology*. Oxford University Press, pp 77–128. (Original work published 1919.)

Wildman WJ, Sosis R (2011) Stability of Groups with Costly Beliefs and Practices. *Journal of Artificial Societies and Social Simulation* 14(3):6. doi:10.18564/jasss.1781.

Data and Code Availability

All simulation code, data, and analysis scripts are available for review: a copy of the model, with its ODD documentation, the frozen calibration parameters, and the committed sweep summaries, is archived at the CoMSES Net Computational Model Library, <https://www.comses.net/codebase-release/9ab5795b-836b-4519-a9a1-16d3d06e400d/>. Upon acceptance the model will be published with a permanent handle and the version-control repository linked.

Supporting Information

Online Resource 1. Supporting text covering the parameter-to-observable mapping, the complete model-parameter tables, the extended model description with the full ODD design-concept narrative and the recorded output fields, classification robustness including the activation-threshold sensitivity, the continuous observability–reward and closure surfaces, the privilege ablation, the endogenous exit-capacity analysis with its anti-circularity test, and computational reproducibility with the committed-run inventory.